Indigenous Place Name Representation in Large Language Models

Rushil Joshi
rjoshi07@umd.edu
University of Maryland
College Park, USA

Ritvik Venkat
rvenkat1@terpmail.umd.edu
University of Maryland
College Park, USA

Nicole R. Schneider
nsch@umd.edu
University of Maryland
College Park, USA

Srivishnu Pyda
spyda@umd.edu
University of Maryland
College Park, USA

Hanan Samet
hjs@cs.umd.edu
University of Maryland
College Park, USA

Abstract

Large language models (LLMs) like ChatGPT and Claude are increasingly used for location-based tasks such as answering geographic questions and generating travel recommendations. However, these models are typically trained on internet-scale data that under-represents rare place names, potentially leading to unequal performance on questions that involve resolving spatial synonyms. This work investigates whether LLMs respond differently to factual queries about dual-named places when given equivalent Indigenous and Anglicized toponyms, using population and area estimates for Australian, Canadian, and New Zealand locations as a case study. In addition to measuring response accuracy, we examine the internal embedding representation of the toponyms, revealing that Indigenous names are more diffusely represented, which correlates to a substantial increase in error in estimating population when places are referred to by their Indigenous names compared to their Anglicized counterparts. We find that through fine-tuning we are able to reduce the disparity and improve performance on questions involving Indigenous place names, suggesting that these biases can be improved even using a small set of data. Our results highlight a representational gap in current LLMs and underscore the need to improve embeddings to enable more culturally equitable LLM performance.

CCS Concepts

• **Computing methodologies** → **Natural language generation**;
• **Information systems** → *Geographic information systems.*

Keywords

fairness, Indigenous and First Nations knowledge, place names, large language models, information access

1 Introduction

In recent years, Large Language Models (LLMs) such as ChatGPT and Claude have become prominent across both technological landscapes and broader society. These models are increasingly used in a range of domains, including education, healthcare, creative writing, language translation, and software development [3, 14]. LLMs are especially popular for spatial tasks, including generating travel plans [20, 23], providing local recommendations [21], and retrieving information with a geographic component [11, 15, 16], such as the history of a cultural landmark [7]. Many of these spatial tasks implicitly involve toponym resolution, the process of matching a text reference mentioning a place name to the actual geographic location of that place in the world [4]. Place name references can be ambiguous, where one name refers to many places [18], or polynymous, where multiple distinct names refer to the same geographic place [12, 13, 19]. LLMs have been shown to perform well at some geospatial questions that involve toponym resolution [1], such as predicting population variables [10], but there is also evidence suggesting that LLMs exhibit biases in certain contexts, particularly when dealing with Indigenous place names [17]. Unlike their Anglicized counterparts, Indigenous toponyms may be less frequently encountered by the models during training, due to an under-representation in internet-based datasets. This disparity in representation may lead to degraded performance or altered behavior in the case of dual-named spatial synonyms when the Indigenous version of the place name is used.

In this paper we address five research questions to better understand the extent and cause of this bias, and mitigate it. The first objective is to evaluate whether LLMs perform differently when presented with geographically factual queries that use Indigenous toponyms versus their Anglicized counterparts. For instance, we compare the model's response to questions like "What is the population of Sydney?" and "What is the population of Warrane?", where the latter uses the Gadigal name for the city instead of the more well-known, Anglicized one. Our experiments reveal that an LLM consistently performs worse when queries includes Indigenous names, often returning significantly less accurate responses, and systematically underestimating the population size, as compared to the performance when the Anglicized counterpart is used in the question.

The second objective of this work is to understand how Anglicized and Indigenous toponyms are represented in the embedding space of an LLM. We find that qualitatively, Indigenous toponyms are more diffusely represented in latent space, which we confirm quantitatively using clustering metrics. This observation suggests the possibility of weaker or less consistent internal modeling of these place names.

The third objective of this work is to determine if the poor performance on geospatial questions involving Indigenous toponyms is correlated to the embedding space representation of place names in the model under evaluation. We conduct a study assessing the

correlation between response accuracy and the embedding space distance between equivalent Indigenous and Anglicized toponyms and find a moderate correlation which further suggests the poor model performance is related to spatially inconsistent internal semantic representations of the toponyms.

The fourth objective of this work is to asses whether the overall performance on factual questions improves after Anglicized-Indigenous equivalency fine-tuning. That is, does fine-tuning the model on place name pairs or relevant facts lead it to answer questions about those places more accurately?

By evaluating the embedding space and performance of two open-source LLMs on a dataset of Indigenous and Anglicized place name questions, we uncover disparities that highlight critical limitations in current LLMs and how they treat Indigenous toponyms. Our findings show that these disparities can be somewhat remedied through fine-tuning, which invites further research on improving embedding representation of Indigenous and other rare place names.

The rest of this paper is organized as follows. In section 2 we survey existing literature on bias and fairness in LLMs, especially with respect to Indigenous knowledge. Next, in section 3 we describe our method for constructing our dataset and fine-tuning models. Then, in section 4 we present our findings, and in section 5 we discuss the significance of our results, before concluding in section 6.

2 Related Work

This section reviews existing literature on geographical bias, name-based biases in embeddings, cultural identity assumptions, and the representation of Indigenous languages in LLMs, highlighting the need for focused analysis on Indigenous toponyms.

Chen et al. [2] investigate whether LLMs develop internal spatial representations that causally influence their outputs. They show that manipulating these internal representations affect the models' performance on tasks requiring geospatial understanding. This suggests that LLMs learn and utilize an internal model of space that is critical to answering spatial questions. Dudy et al. [5] builds off of this work by examining biases in LLM-generated geographical recommendations, revealing a tendency to favor urban and economically advantageous areas in the U.S. for relocation, tourism, and business queries. Their study demonstrates that LLMs often under-represent less-documented regions, which perpetuates existing socio-economic disparities.

Hu et al. [9] demonstrates that LLMs also exhibit social identity biases, mirroring human tendencies such as ingroup favoritism and outgroup derogation. This social mirroring effect is especially relevant when considering how LLMs might process cultural inputs such as Indigenous versus Anglicized place names. Since societal familiarity and dominant cultural norms often center around Anglicized terminology, LLMs may implicitly favor these variants in both recognition and response quality. If language models reflect societal tendencies to privilege dominant group identifiers, it follows that they may under-represent or deprioritize Indigenous toponyms that generally fall outside of mainstream exposure.

Further, Wiechetek et al. [22] reveals that LLMs make incorrect assumptions in language generation and machine translation tasks involving Indigenous languages. They posit these issues are caused by a lack of training data and a lack of expert knowledge when using data harvested from the web to train LLMs. Their work motivates the need to examine the internal representations in LLMs to better understand performance discrepancies on questions involving Indigenous language. Taken together, these studies expose various layers of bias in how LLMs process place-based and Indigenous-related queries. We add to the current landscape but investigating how LLMs represent Indigenous place names, which invites further research towards improving how LLMs treat Indigenous and Anglicized spatial synonyms.

3 Method

This section outlines the method we use to investigate performance disparities and embedding characteristics of Indigenous versus Anglicized toponyms.

3.1 Research Questions

RQ1. Does performance differ based on use of an Anglicized versus an Indigenous toponym? To evaluate RQ1, we query a model with pairs of toponyms, asking the same population question for each one and recording how the output varies.

RQ2. Does an LLM internally represent Anglicized and Indigenous toponyms differently? To answer RQ2, we extract embeddings from the model and analyze their characteristics for Anglicized and Indigenous toponyms.

RQ3. Do differences in performance across Anglicized and Indigenous toponyms correlate to their positioning in embedding space? RQ3 aims to quantify whether spatial synonyms that are represented farther apart from each other in embedding space are also harder for the model to answer questions about. To address this question, we measure the correlation between embedding space distance for Anglicized-Indigenous toponym pairs and model error on the corresponding questions about those toponyms.

RQ4. Does fine-tuning the model on relevant spatial synonyms improve performance on factual questions involving those toponyms? RQ5 aims to determine if the model's knowledge of facts about places can be improved by fine-tuning it on place name mappings, allowing it to associate multiple valid names for the same place.

3.2 Dataset Construction

We construct a custom dataset [1] consisting of paired Indigenous and Anglicized toponyms across Australia, Canada, and New Zealand. The dataset includes 102 toponyms (51 pairs). Each entry in the dataset includes (i) the Indigenous place name, (ii) its Anglicized counterpart, (iii) 2 factual questions about the population and size of the place, and (iv) 2 hand-labeled ground truth responses. For this study, the factual questions are formatted as

(1) "What was the population of `[location]`, `[country]` in 2021?"
(2) "What was the area of `[location]`, `[country]` in 2021?"

where the `[location]` placeholder is replaced by either the Indigenous or Anglicized version of a place name, and the `[country]` placeholder is replaced by the country containing that place. We

[1] https://github.com/nicoleschneider/LLM-Embeddings

Method	Question Type	Toponym Type	Model	R-Acc (↑)	ER (↑)	A-Acc (↑)
Baseline	Population	Indigenous	llama-3.2b	0.3333	-0.0577	0.8846
			llama-8b	0.1818	-0.6364	0.1818
		Anglicized	llama-3.2b	0.7143	0.3462	0.7692
			llama-8b	0.6667	0.2727	0.7273
	Area	Indigenous	llama-3.2b	0.1250	-0.1154	0.8654
			llama-8b	0.0909	-0.8182	0.0909
		Anglicized	llama-3.2b	0.3000	-0.3077	0.4615
			llama-8b	0.4545	-0.0909	0.4545
Fine-tuned (Pairs)	Population	Indigenous	llama-3.2b	0.3913	-0.0962	0.7308
			llama-8b	0.1818	-0.6364	0.1818
		Anglicized	llama-3.2b	0.7059	0.4038	0.7115
			llama-8b	0.6364	0.2727	0.6364
	Area	Indigenous	llama-3.2b	0.1316	-0.5385	0.3654
			llama-8b	0.0909	-0.8182	0.0909
		Anglicized	llama-3.2b	0.2745	-0.4423	0.2885
			llama-8b	0.4545	-0.0909	0.4545
Fine-tuned (Facts)	Population	Indigenous	llama-3.2b	0.5000	0.0000	0.8462
			llama-8b	0.2727	-0.4545	0.2727
		Anglicized	llama-3.2b	0.7400	0.4615	0.7500
			llama-8b	0.7273	-0.4545	0.2727
	Area	Indigenous	llama-3.2b	0.0455	-0.3846	0.5962
			llama-8b	0.0909	-0.8182	0.0909
		Anglicized	llama-3.2b	0.1915	-0.5577	0.2692
			llama-8b	0.3636	-0.2727	0.3636

Table 1: Performance results on test questions for baseline and fine-tuned models.

sourced the ground truth population values for Australia from the Australian Bureau of Statistics [2]. We sourced the ground truth population values for Canada from the Canadian Government Census Page [3]. We sourced the ground truth population values for New Zealand from the New Zealand Statistics Page [4]. The dataset statistics are summarized in Table 2.

Country	Name	Toponyms	Fine-tuning Questions	Test Questions
Australia	Indigenous	31	93	62
	Anglicized	31	93	62
Canada	Indigenous	10	30	20
	Anglicized	10	30	20
New Zealand	Indigenous	10	30	20
	Anglicized	10	30	20
Total		102	306	204

Table 2: Dataset statistics.

We do an 80/20 train/test split by city, with held out data points used for the results reported.

[2]https://www.abs.gov.au/
[3]https://www12.statcan.gc.ca/census-recensement/index-eng.cfm
[4]https://www.stats.govt.nz/topics/environment/

3.3 Model

We use Meta's `llama3-3.2b` and `llama3-8b` model as a case study to evaluate performance on Anglicized and Indigenous toponyms and characterize the embedding space representation of these toponyms.

3.4 Fine-Tuning

To fine tune the place name representations in the model, we formulate question and answer pairs about the alternate place names from the Anglicized and Indigenous place name labels in our dataset.

For each place name pair we construct 3 name equivalency question-response pairs for fine tuning:

(1) "What is the alternate name for the place `[location]` in `[country]`?"
(2) "What is the `[Indigenous/Anglicised]` name for `[location]` in `[country]`?"
(3) "In `[country]`, what is another name for `[location]`?"

For each of these questions the correct response is the name of the alternate (Indigenous or Anglicized) version of the place name provided as `[location]`.

We also include a fine-tuning setting in which fact questions and correct responses are used for tuning, similar to the type of question used in our evaluation, covering population and area of different places. These questions are formulated as

(1) "What is the population of `[location]` in `[country]`?"
(2) "What is the area of `[location]` in `[country]`?"

The goal of the fine tuning is to teach the model to answer questions regardless of whether the Indigenous or Anglicized version of the place name is used, since it should associate these different namings of the same place as related.

We use Low-Rank Adaptation (LoRA) [8] for the fine-tuning approach. LoRA freezes the original weights and learns a low-rank update for selected linear layers. For a pretrained weight matrix $W_0 \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$, the update is

$$W = W_0 + \Delta W, \qquad \Delta W = \frac{\alpha}{r} BA, \tag{1}$$

where $A \in \mathbb{R}^{r \times d_{\text{in}}}$ and $B \in \mathbb{R}^{d_{\text{out}} \times r}$ are trainable matrices, $r \ll \min(d_{\text{in}}, d_{\text{out}})$ is the LoRA rank, and α is a scaling factor. During training, only A and B are updated, while W_0 remains fixed. The modified linear transformation is therefore

$$h = W_0 x + \frac{\alpha}{r} BAx. \tag{2}$$

After training, the learned update ΔW is merged into the base weight matrix for inference.

3.5 Evaluation Metrics

RQ1. Since LLMs tend to abstain from answering questions about Indigenous toponyms, we carefully choose evaluation metrics that account for abstentions appropriately. We evaluate correctness of answers on three axes, which are defined below. In order to use these metrics, we bin responses into correct, incorrect, and abstain, which requires setting a threshold for correctness. For our evaluations we choose 20% as the threshold for classifying a response as correct.

We follow Feng et al. [6] and evaluate correctness of LLM responses against ground truth using the following accuracy metrics, which account for correct, incorrect, and abstain responses, which we represent using the symbols outlined below.

	Correct	Incorrect
Answered	$\mathcal{A}$	$\mathcal{C}$
Abstained	$\mathcal{B}$	$\mathcal{D}$

Reliable Accuracy (R-Acc), measured as

$$\text{R-Acc} = \frac{\mathcal{A}}{\mathcal{A} + \mathcal{C}} \tag{3}$$

which indicates to what extent LLM-generated answers (not abstentions) could be trusted (i.e., out of all questions answered, how many are correct?) [6].

Effective Reliability (ER), measured as

$$\text{ER} = \frac{\mathcal{A} - \mathcal{C}}{\mathcal{A} + \mathcal{B} + \mathcal{C} + \mathcal{D}} \tag{4}$$

which strikes a balance between reliability and coverage (i.e., out of all questions, how many more are answered correctly than incorrectly?) [6]

Abstention accuracy (A-Acc), measured as

$$\text{A-Acc} = \frac{\mathcal{A} + \mathcal{D}}{\mathcal{A} + \mathcal{B} + \mathcal{C} + \mathcal{D}} \tag{5}$$

which evaluates whether the abstain decisions are correct (i.e. did the LLM abstain when it would provide an incorrect answer and vice versa?) [6].

This evaluation is performed separately for questions using Indigenous names and those using Anglicized names, allowing for a quantitative comparison of model performance across the two naming conventions.

RQ2. We assess the embedding space by projecting the model's high-dimensional embeddings for the place names in our dataset into two dimensions using t-distributed Stochastic Neighbor Embedding (t-SNE), a dimensionality reduction technique designed to preserve local structure. The resulting 2D visualization is used to qualitatively assess whether Indigenous and Anglicized names form distinct visual clusters or are interspersed.

RQ3. We assess the correlation between performance and embedding space position by comparing the embedding space distances for a toponym pair to the model performance error on the corresponding Indigenous toponyms. To measure embedding space distance, we employ Euclidean distance

$$d_E(A, B) = ||A - B||$$

to obtain a vector D of embedding space distances for the different toponym pairs. To measure the model performance, for each Indigenous toponym we compute

$$p_\delta = |P - \hat{P}_{Model}|,$$

where P is ground truth population and $\hat{P}_{Model}$ is population estimated by the model, and construct a vector P_Δ representing the p_δ values for the entire dataset. To measure correlation, we use the Pearson correlation coefficient [5] which is computed as

$$r = \frac{\sum (P_\Delta - m_{P_\Delta})(D - m_D)}{\sqrt{\sum (P_\Delta - m_{P_\Delta})^2 \sum (D - m_D)^2}}$$

where m_{P_Δ} and m_D are the means of vectors P_Δ and D, respectively.

RQ4. We evaluate the embedding space following the evaluation protocol for *RQ3*, applying the same test and evaluation after fine-tuning the models.

4 Results

We present model performance, embedding space analysis, and correlation between embedding position and model error.

4.1 RQ1: Performance Disparities

To evaluate accuracy, we query the model with 11 held out population and area questions from our dataset, recording performance on Indigenous and Anglicized toponyms in Table 1. We find that accuracy metrics for the Indigenous toponym questions are significantly worse than for the Anglicized ones. This reflects a general trend that model performance degrades when Indigenous place names are used, since they appear less frequently in training data and are less likely to be recognized by the model. Investigating further, we find systematic underestimation of the populations when the Indigenous toponym is used in the prompt, and further find that the population values returned for Indigenous toponyms were statistically significantly [6] smaller than those returned for Anglicized toponyms. In some cases, like for Sydney/Warrane, the model

[5] https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.pearsonr.html
[6] Using a paired t-Test with $\alpha = 0.1$

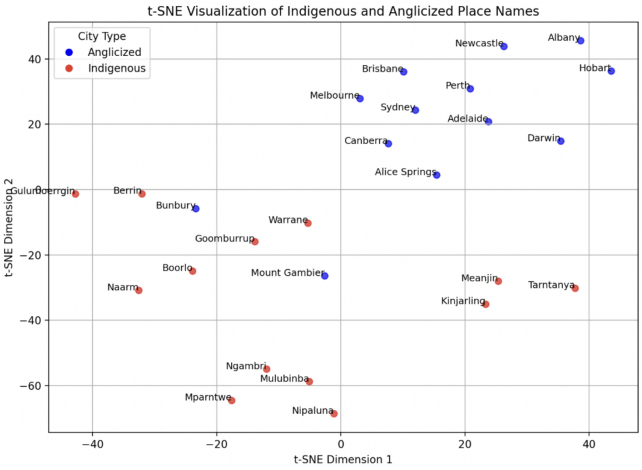


Figure 1: A t-SNE visualization of `llama3-8b` embeddings for Anglicized and Indigenous toponyms located i nAustralia. Anglicized embeddings appear more clustered, while Indigenous embeddings are more scattered.

returned a near-correct population for the Anglicized toponym (which is "Sydney") but it underestimated the population of the Indigenous synonym (which is "Warrane") by more than an order of magnitude, despite both names referring to approximately the same place.

4.2 RQ2: Embedding Space Characteristics

To investigate how the model internally represents toponyms, we extract the embeddings for the same toponyms tested in RQ1. We then project the embeddings into two dimensions using t-distributed Stochastic Neighbor Embedding (t-SNE) to enable visual examination of their structure (Figure 1). Anglicized toponyms appear clustered in the embedding space, whereas their Indigenous versions appear scattered, and are not paired with their Anglicized counterparts, as one might expect if the model represented the spatial synonyms with similar meaning. In context of previous work by Chen et al. [2], which demonstrated that LLMs develop internal spatial representations of place, our findings suggest that this spatial modeling does not extend to Indigenous toponyms.

4.3 RQ3: Correlation Between Performance and Embedding Space

To understand the relationship between toponym representation and model performance, we examine the correlation between embedding distances and model errors on the Indigenous toponyms. [7] We find a moderate correlation of 0.29, indicating model performance tends to be worse when an Indigenous toponym embedding vector is far from its corresponding anglicized embedding vector. This finding indicates a potentially interesting line of future work, where model embeddings for toponyms, especially rare ones, can be trained with an additional objective, to improve their consistency with the spatial reality they represent, which may in turn improve model performance on questions pertaining to those places.

[7] One degenerate data point is omitted from the analysis

4.4 RQ4: Effect of fine-tuning on Question Answer Performance

Results for both fine-tuning approaches (place name pairs and full facts) are shown in table 1 and visualized in figure 3. Fine tuning the model on place name pairs did not substantially improve performance across any of the metrics. Fine-tuning the model using facts about population or area did improve performance, especially in terms of reliability accuracy and abstention accuracy. Notably, `llama3-8b` performs worse on Indigenous place name questions out of the box than `llama3-3.2b`, even after fine-tuning.

5 Discussion

We find clear disparities in how an LLM responds to queries involving Indigenous toponyms compared to their Anglicized counterparts, as well as how those toponyms are represented in the model's internal embedding space. For Anglicized toponyms, the models appear to have constructed coherent semantic clusters. In contrast, the scattered nature of Indigenous embeddings reflects an incomplete or poorly integrated spatial model for these culturally significant place names. Further, we find a moderate correlation between model performance on an Indigenous toponym and the embedding distance from that toponym to its Anglicized counterpart.

By fine-tuning the models on geographic facts about Indigenous and Anglicized place names, performance improves. We find that fine-tuning on the place name mappings from Indigenous to Anglicized version does not improve performance, indicating the context of the place is important to the LLM in order to recognize and answer correctly about the place.

Our results point to systemic issues in how LLMs represent Indigenous language and geographic information and reinforce prior observations in the literature that Indigenous terms are underrepresented in training data [22], and that current models often fall short in capturing culturally specific knowledge. For individuals using Indigenous toponyms in queries, the model provides substantially worse responses, which underscores the need for more representative pretraining data, benchmarks that capture Indigenous representation, and additional training objectives that seek to make model embeddings more spatially consistent. Our fine-tuning approach shows that it is possible to improve performance, and lessen the gap between performance on Indigenous and Anglicized place names, but it raises an important question about biases in training data, and how to avoid this gap in the first place.

6 Conclusion

This work demonstrates that large language models exhibit clear disparities in how they represent and respond to Indigenous versus Anglicized place names. Through empirical analysis of response accuracy, embedding coherence, and correlational analysis, we show that Indigenous toponyms are less consistently represented and more prone to model error. These findings highlight a structural bias in current LLMs likely rooted in imbalanced training data. We find that through fine-tuning we are able to reduce the disparity

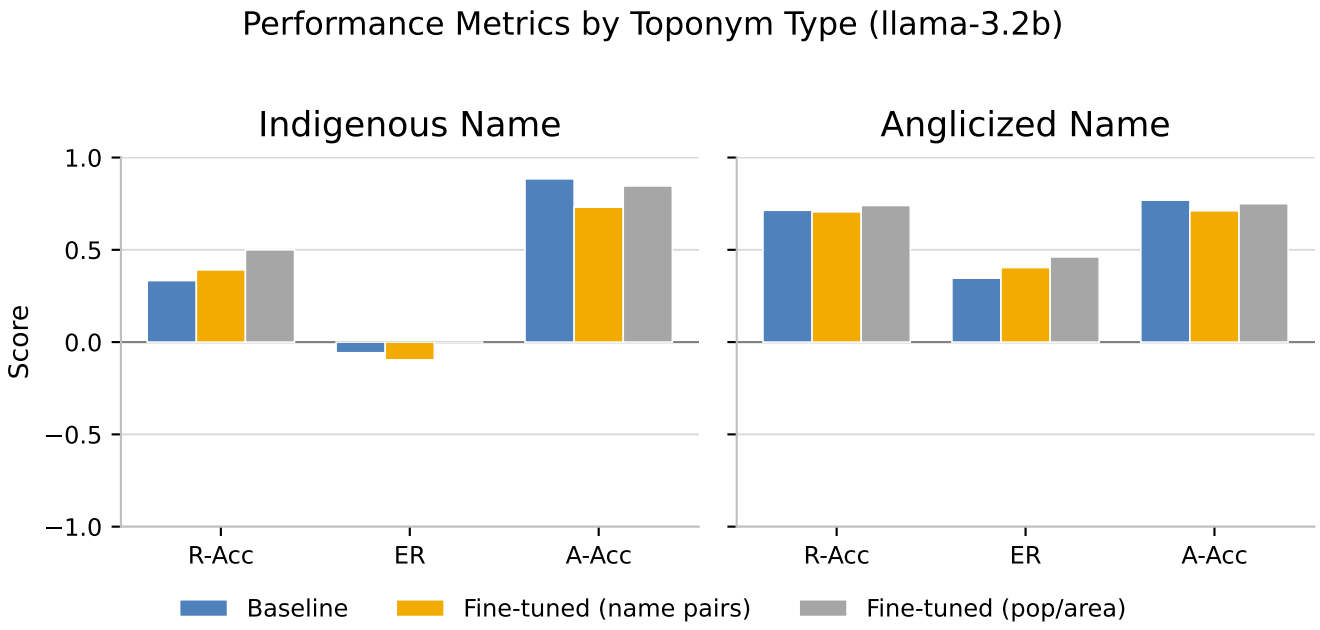


Figure 2: Performance results of `llama3-3.2b` on all test questions.

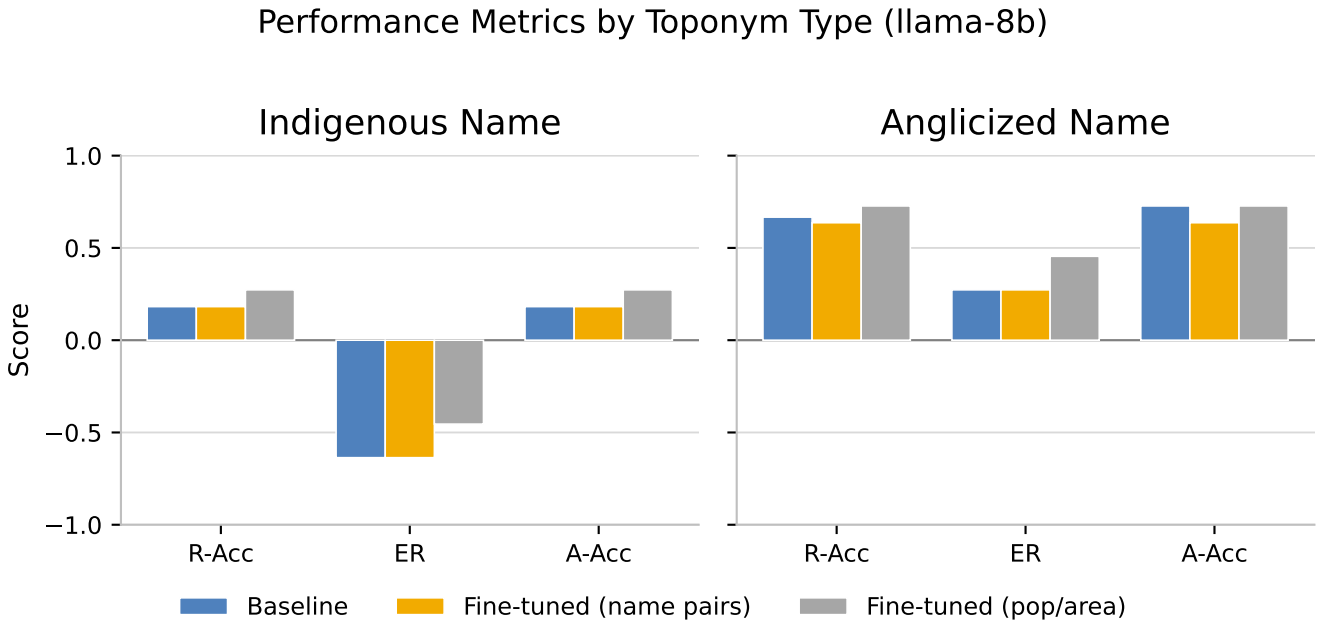


Figure 3: Performance results of `llama3-8b` on all test questions.

and improve performance on questions involving Indigenous place names. By uncovering these representational gaps, this study underscores limitations in current LLMs and contributes a focused framework for evaluating geographic and cultural bias in language models, laying the groundwork for future research towards more equitable and culturally aware language models.

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